

Fine-Gray Subdistribution Hazards

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Introduction

In competing-risks data, two questions often co-exist: “what is this covariate’s mechanistic effect on the event of interest” and “what is its effect on the population-level probability of experiencing that event by time t ?” Cause-specific Cox regression answers the first; Fine-Gray subdistribution-hazards regression answers the second. The two questions can disagree dramatically — a covariate can leave the cause-specific hazard unchanged while still raising or lowering the cumulative incidence because it changes the competing-event hazard. Reporting only one when both are relevant is a common analytical mistake.

Prerequisites

A working understanding of competing-risks data, cumulative incidence functions, and the cause-specific vs subdistribution-hazards distinction.

Theory

The subdistribution hazard for cause 1 is

$$h_1^{\text{sub}}(t | x) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t, C = 1 | T \geq t \text{ or } (T < t \text{ and } C \neq 1), x)}{\Delta t}.$$

The risk set keeps subjects who have already experienced a competing event, with diminishing weights. Under the Fine-Gray model

$$h_1^{\text{sub}}(t | x) = h_{10}(t) \exp(x^\top \beta),$$

the exponentiated coefficient $\exp(\beta_j)$ is a *subdistribution hazard ratio* and maps directly onto changes in cumulative incidence: a SHR > 1 means higher cumulative incidence of cause 1 per unit of covariate. The model is fitted by a modified partial likelihood with inverse-probability-of-censoring weights.

Assumptions

Proportional subdistribution hazards (an analogue of proportional cause-specific hazards), independent censoring (or correctly weighted), and accurate cause classification.

R Implementation

```
library(cmprsk)

set.seed(2026)
t_time <- rexp(300, 0.3)
cause <- sample(0:2, 300, replace = TRUE, prob = c(0.3, 0.5, 0.2))
x1 <- rnorm(300); x2 <- rbinom(300, 1, 0.5)

fg <- crr(ftime = t_time, fstatus = cause, cov1 = cbind(x1, x2),
```

```
failcode = 1, cencode = 0)
summary(fg)
```

Output & Results

`crr()` returns subdistribution coefficients on the log-sHR scale, standard errors, p -values, and (after `summary()`) exponentiated subdistribution hazard ratios with confidence intervals. The output structure matches Cox regression but the interpretation refers to cumulative incidence rather than cause-specific hazard.

Interpretation

A reporting sentence: “Fine-Gray subdistribution-hazards regression for cause 1 estimated sHR = 1.45 for x_1 (95 % CI 1.10 to 1.90), meaning a 45 % higher cumulative incidence of cause 1 per unit increase in x_1 ; a cause-specific Cox model on the same data gave HR 1.32, indicating that part of the effect on cumulative incidence comes via the competing-event hazard.” Always report both subdistribution and cause-specific HRs when they differ; they answer different questions.

Practical Tips

- Subdistribution hazard ratios do *not* equal cause-specific hazard ratios; treating them as interchangeable is a common error.
- Use cause-specific Cox for *etiologic* questions (“what affects the disease mechanism?”); use Fine-Gray for *prognostic* / prediction questions (“what affects the probability of disease over time?”).
- For prediction tools (nomograms, risk calculators), Fine-Gray is the correct framework because absolute risk depends on cumulative incidence.
- Report both approaches when both questions matter; transparency about the difference prevents readers from misinterpreting one as the other.
- `riskRegression::FGR()` offers a modern alternative with cleaner diagnostics and better small-sample handling.
- Check proportional subdistribution hazards via `crskdiag` or simulated residual plots; the assumption can fail even when proportional cause-specific hazards holds.

R Packages Used

`cmprsk` for `crr()` and `cuminc()`; `riskRegression` for `FGR()`, `CSC()` (cause-specific Cox), and unified prediction interfaces; `tidycmprsk` for tidy-style competing-risks workflows; `survival` for Cox regression on cause-specific hazards as the complementary fit.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation